

Basic Information

Address Wing Howard To, Ph.D.
5621 Charlotte Way, APT8
Livermore, CA 94550

Email **wto@csustan.edu**

Mobile (805)-455-3373

Linkedin <https://www.linkedin.com/in/wing-to/>

Github <https://wtoslac.github.io/>

Education

- 2006-2012 **Doctorate of Philosophy in Physics (Experimental Particle Physics)**, *Department of Physics*, University of California, Santa Barbara, CA, USA
- 2005-2006 **Bachelor of Science in Physics (Experimental Quantum Optics)**, *Department of Physics*, University of California, Santa Barbara, CA, USA
- 2002-2004 **Associate of Science in Natural Sciences**, Citrus Community College, Glendora, CA, USA

Academic Appointments and Affiliations

- 2017-Current **Assistant & Associate Professor**, *Department of Physics, Physical Sciences and Geology*
California State University, Stanislaus, Turlock, CA, USA
- 2024-Current **Research Affiliate**, *Earth Science Division & Bay Area Environmental Research Institute*
NASA Ames Research Center, Moffett Field, CA, USA
- 2022-Current **Research Affiliate**, *Nuclear Sciences Division & Physics Division*
Lawrence Berkeley National Lab, Berkeley, CA, USA
- 2017-2021 **Assistant Professor**, *Department of Physics, Physical Sciences and Geology*
California State University, Stanislaus, Turlock, CA, USA
- 2017-2022 **Research Affiliate**, *Kavli Institute for Particle Astrophysics and Cosmology (KIPAC)*
Stanford University & SLAC National Accelerator Lab, Menlo Park, CA, USA
- 2014-2017 **Research Associate**, *Kavli Institute for Particle Astrophysics and Cosmology (KIPAC)*
Stanford University & SLAC National Accelerator Lab, Menlo Park, CA, USA
- 2013-2014 **Postdoctoral Scholar** *Department of Physics*
Case Western Reserve University, Cleveland, OH, USA

Service and Committee Contribution

- 2021-Current **Chair** of Physics & Physical Sciences Program Curriculum Committee
- 2021-Current **Member** of University Curriculum Committee
- 2022-2025 **Member** of University Research, Scholarly, Creative Activities Policy Committee
- 2022-2024 **Chair** of Department of Physics, Retention, Promotion and Tenure Committee
- 2022-2023 **Member** of College of Science Dean Search Committee
- 2019-2025 **Member** of Steering Committee on Department of Education Grant: *Students Transitioning to Engaged and Motivated (STEM) Success*, Grant#P031C160070
- 2018-2023 **Member** of Steering Committee of the Cal-Bridge Program. <https://calbridge.org/>
- 2019-2021 **Coordinator** STEM Discovery Academy for Incoming Freshman and Transfer Students
- 2017-2022 **Faculty Advisor** for Society of Physics Students Local Chapter

Grants Awarded and Donations

- Spring 2025 Recipient of \$36,000 in Donation of Electronics Equipment from Keysight Technologies.
- 2024-2026 PI on NASA ROSES-23, Science Mission Directorate Bridge Grant, *Fire & Air: Burning Issues in the Central Valley: Unraveling Fire's influence on Air Quality, Fuel Mapping, and Carbon Dynamics*, \$355k, Award# 80NSSC24K1065
- 2023-2026 PI on Department of Energy, RENEW-HEP Grant, *Growth and Research Opportunities With Traineeships in High energy physics at Minority Serving Institutions (GROWTH-MSI)*, \$950k, Award# DE-SC0023725
- 2023-2024 PI on Lawrence Berkeley Lab Subaward on National Nuclear Security Agency Grant, *Collaboration between LBL and Stan State on the Eos Experiment*: \$80k, Subcontract #7677697
- 2022-2027 PI on NASA ROSES-2021 Grant: *Increasing Diversity in Earth and Atmospheric Sciences with Courses, Outreach and Research Programs (IDEAS-CORP) at Minority Serving Institutions*, \$520k Award # 80NSSC22K1782
- 2019-2022 Co-PI on NSF HSI Grant: *Building Capacity: Collaboration for Inclusive and Engaging Curriculum, Instruction, and Achievement* \$1.5M Award # 1832558
- 2021-2022 PI: Innovate, Design, Excel and Assess for Success (IDEAS) Grant: *Warrior Team Science Internship for Studying the Environmental Effect of California Wildfires*, Award \$5000
- 2021-2022 PI: Research, Scholarly and Creative Activities (RSCA) Grant: *Research to Study the Effects of Wildfires on the Environment and Atmosphere*, Award \$8400
- 2020-2021 PI: Research, Scholarly and Creative Activities (RSCA) Grant: *Seeding an Environmental Science research group at Stanislaus State with on atmospheric effects of California wildfires*, Award \$7200
- 2018-2019 PI: Research, Scholarly and Creative Activities (RSCA) Grant: *Undergraduate Research Experience in Experimental Dark Matter Detector in South Dakota*, Award \$5600
- 2017-2018 PI: Research, Scholarly and Creative Activities (RSCA) Grant: *Research Experience for Undergraduate in Dark Matter Experiments*, Award \$5000

Publication List

- Aug 2023 J. Lee, J.D. Mirocha, N. Lareau, T. Whitney, **W. H. To**, A. Kochanski, W. Lassman, *Sensitivity of pyrocumulus convection to tree mortality during the 2020 Creek fire in California*, Geophysical Research Letters, DOI: 10.1029/2023GL104193.
- Feb 2023 T. Anderson, ..., **W.H. To**, et al., *EOS: conceptual design for a demonstrator for hybrid optical detector technology*, Journal of Instrumentation, Volume 18, DOI 10.1088/1748-0221/18/02/P02009
- Oct 2022 LUX Collaboration: S. Kravitz, ..., **W.H. To**, et al., *Fast and flexible analysis of direct dark matter search data with machine learning*, Phys. Rev. D 106, 072009, arXiv:2201.05734
- Sep 2021 LUX Collaboration: G. Rischbieter, ..., **W.H. To**, et al., *Constraints on effective field theory couplings using 311.2 days of LUX data*, Phys. Rev. D 104, 062005, arXiv:2102.06998
- Dec 2020 LUX Collaboration: **W.H. To**, ..., V. Velan, et al., *Discrimination of electronic recoils from nuclear recoils in two-phase xenon time projection chambers*, Phys. Rev. D 102, 112002, arXiv:2004.06304
- Nov 2020 LUX Collaboration: **W.H. To**, ..., J. Xu, et al., *Investigation of background electron emission in the LUX detector*, Phys. Rev. D 102, 092004, arXiv:2004.07791
- Jul 2019 LUX Collaboration: J. Balajthy, ..., **W.H. To**, et al., *Improved measurements of the beta-decay response of liquid xenon with the LUX detector*, Phys. Rev. D 100, 022002, arXiv:1903.12372
- Jun 2018 LUX Collaboration: C. Carmona, ..., **W.H. To**, et al., *Calibration, event reconstruction, data analysis and limits calculation for the LUX dark matter experiment*, Phys. Rev. D 97, 102008, arXiv:1712.05696
- Jan 2018 D.S. Akerib, ..., C. Lee, **W.H. To**, *Chromatographic separation of radioactive noble gases from Xenon*, Astroparticle Physics, Volume 97, Pages 80-87, <https://doi.org/10.1016/j.astropartphys.2017.10.014>
- Jul 2017 LUX Collaboration: C. Nehr Korn, ..., **W.H. To**, et al., *Limits on spin-dependent WIMP-nucleon cross section obtained from the complete LUX exposure*, Phys. Rev. Lett. 118, 251302, arXiv:1705.03380
- Jan 2017 LUX Collaboration: A. Manalaysay, ..., **W.H. To**, et al., *Results from a Search for Dark Matter in the Complete LUX Exposure*, Phys. Rev. Lett. 118, 021303, arXiv:1608.7648